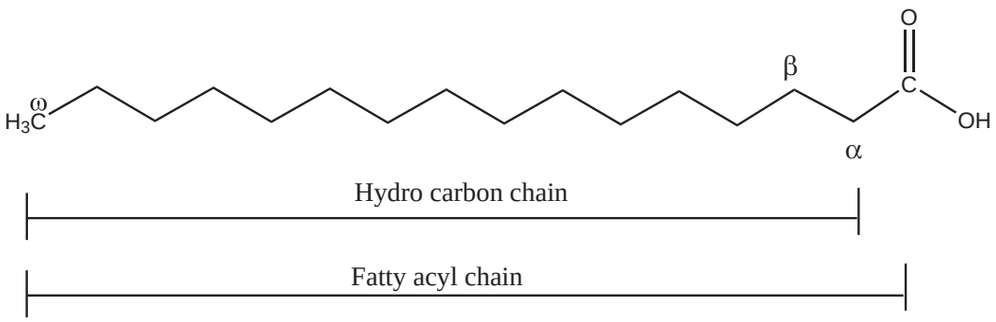
vi. Apart from the general functions lipids serve as pigments (carotene), hormones, signaling molecules (eicosanoids), cofactors (vitamin K), detergents (bile salts) and many other functions.

5.2 BIOSYNTHESIS OF FATTY ACIDS

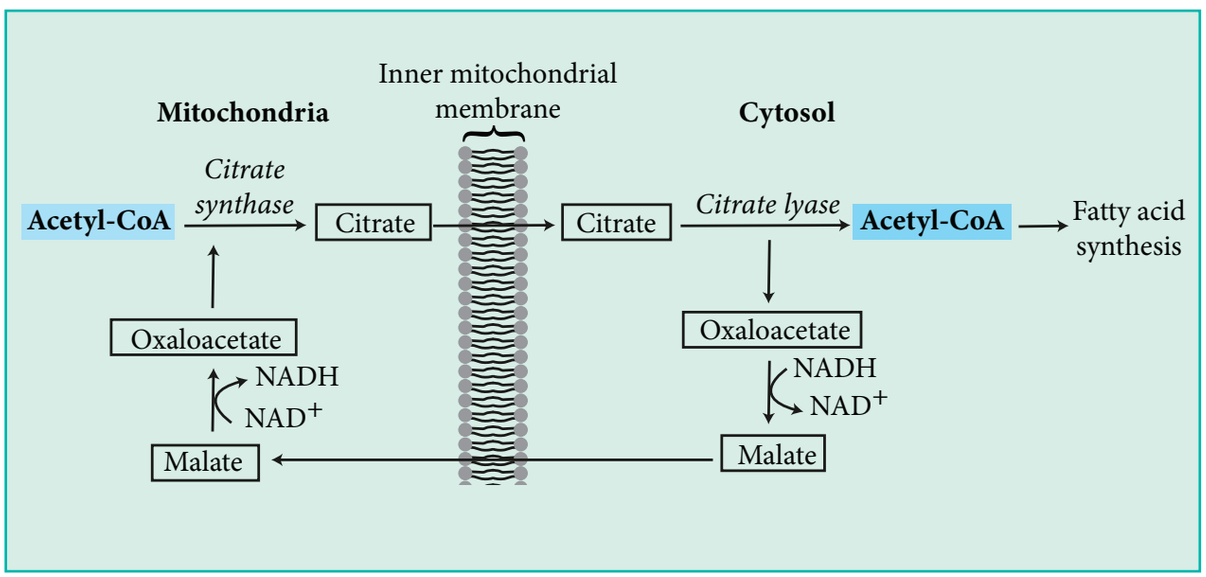
Fatty acids are the simplest form of lipids and they serve as the constituent in a large number of complex form of lipids. They are long chain hydrocarbons (4 to 36 carbons long) with one carboxyl group. Fatty acids are amphipathic in nature; that is, they have both polar and non-polar ends.



In a fatty acid the second carbon is referred as α carbon, third carbon as β carbon and the methyl carbon is called as ω carbon. The alkyl chain in fatty acids may be saturated or unsaturated.

Biosynthesis of fatty acids in animals occurs mainly in the cytosol of adipose tissue, mammary glands, and liver. In case of plants, it occurs in the stroma of chloroplast.

Biosynthesis of fatty acids involves stepwise addition of 2 carbon units, which is supplied by Acetyl CoA. Acetyl CoA is a product of catabolism of carbohydrates, fatty acids and many amino acids. Acetyl CoA formed in the matrix of mitochondria cannot cross the inner mitochondrial membrane and a shuttle system is required to transport the 2-carbon units out of the mitochondria into the cytosol. This is called Acetyl CoA shuttle system.



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